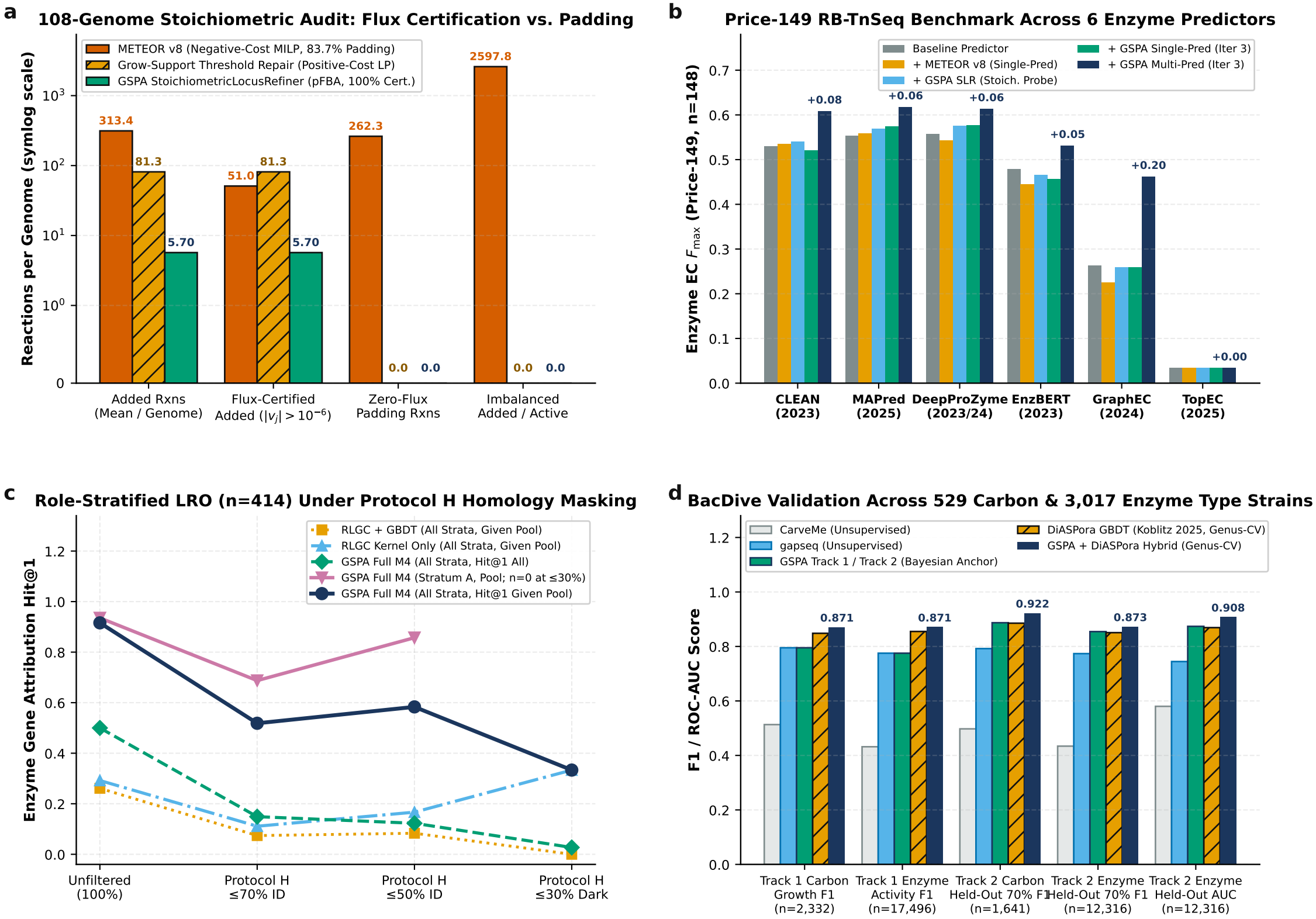
a
108-Genome Stoichiometric Audit: Flux Certification vs. Padding
METEOR v8 (Negative-Cost MILP, 83.7% Padding)
Grow-Support Threshold Repair (Positive-Cost LP)
GSPA StoichiometricLocusRefiner (pFBA, 100% Cert.)
Reactions per Genome (symlog scale)
313.4
81.3
5.70
51.0
81.3
5.70
262.3
0.0
0.0
2597.8
0.0
0.0
Added Rxns (Mean / Genome)
Flux-Certified Added ($|v_j| > 10^{-6}$)
Zero-Flux Padding Rxns
Imbalanced Added / Active
b
Price-149 RB-TnSeq Benchmark Across 6 Enzyme Predictors
Baseline Predictor
+ METEOR v8 (Single-Pred)
+ GSPA SLR (Stoich. Probe)
+ GSPA Single-Pred (Iter 3)
+ GSPA Multi-Pred (Iter 3)
Enzyme EC F_{max} (Price-149, n=148)
+0.08
+0.06
+0.06
+0.05
+0.20
+0.00
CLEAN (2023)
MAPred (2025)
DeepProZyme (2023/24)
EnzBERT (2023)
GraphEC (2024)
TopEC (2025)
c
Role-Stratified LRO (n=414) Under Protocol H Homology Masking
RLGC + GBDT (All Strata, Given Pool)
RLGC Kernel Only (All Strata, Given Pool)
GSPA Full M4 (All Strata, Hit@1 All)
GSPA Full M4 (Stratum A, Pool; n=0 at ≤30%)
GSPA Full M4 (All Strata, Hit@1 Given Pool)
Enzyme Gene Attribution Hit@1
Unfiltered (100%)
Protocol H ≤70% ID
Protocol H ≤50% ID
Protocol H ≤30% Dark
d
BacDive Validation Across 529 Carbon & 3,017 Enzyme Type Strains
CarveMe (Unsupervised)
gapseq (Unsupervised)
GSPA Track 1 / Track 2 (Bayesian Anchor)
DiASPora GBDT (Koblitz 2025, Genus-CV)
GSPA + DiASPora Hybrid (Genus-CV)
F1 / ROC-AUC Score
0.871
0.871
0.922
0.873
0.908
Track 1 Carbon Growth F1 (n=2,332)
Track 1 Enzyme Activity F1 (n=17,496)
Track 2 Carbon Held-Out 70% F1 (n=1,641)
Track 2 Enzyme Held-Out 70% F1 (n=12,316)
Track 2 Enzyme Held-Out AUC (n=12,316)